CSE 225 Data Structures
Week #8

**B**
Quiz #3

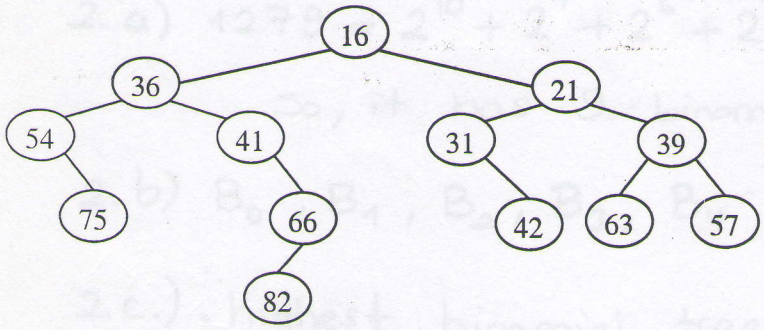
Borahan Tümer, Ph.D.
Fall 2006

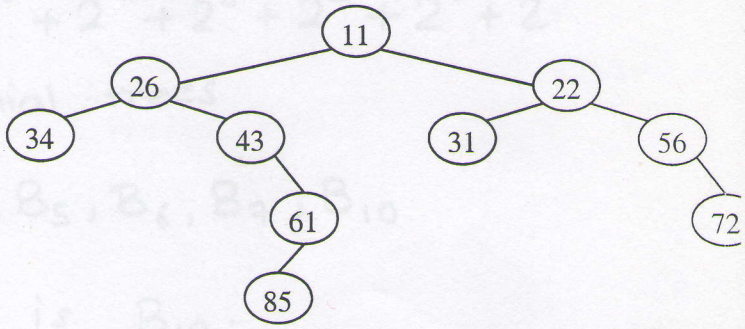
## Q1.

Two heaps are given below.

a) Are they leftist heaps?

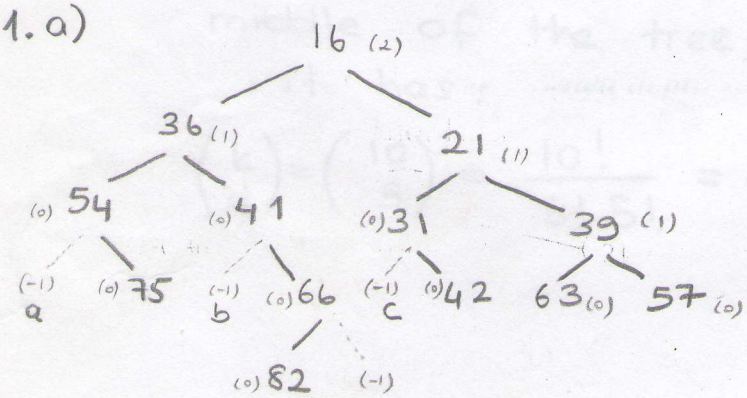
b) Merge them ***recursively*** in ***O(lg n)*** time where ***n*** is the number of elements in both heaps! Make any (or both if necessary) in case they are not LHs!

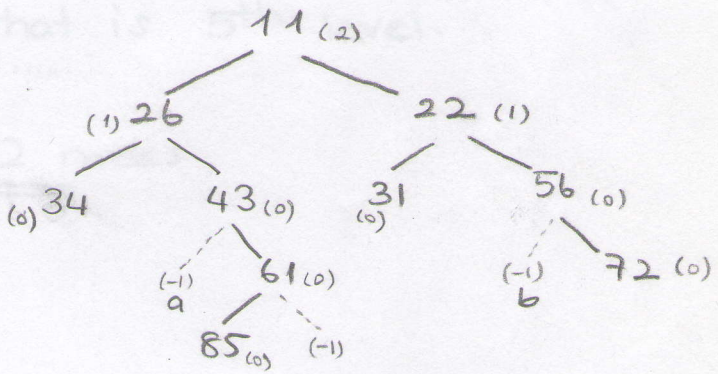



Npl : length of the shortest path from X to a node without two children

1.a)

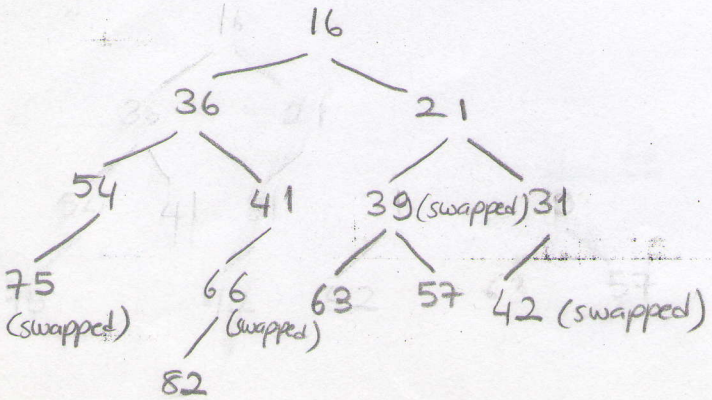


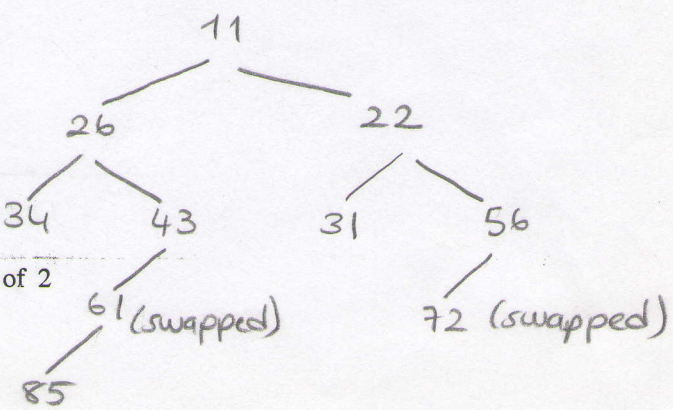




Not leftist : 1) Npl (31) < Npl (39)
2) Npl (a) < Npl (75)
3) Npl (b) < Npl (66)
4) Npl (c) < Npl (42)

Not leftist : 1) Npl (a) < Npl (61)
2) Npl (b) < Npl (72)

1.b) First, make the heaps leftist

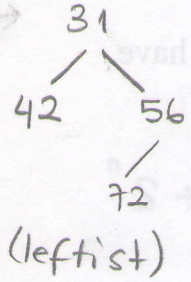


1.b) cont. Then, merge the heaps. (B)

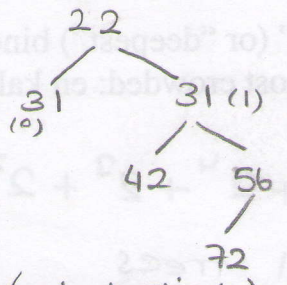
Comparison table: 16 - 11 (5)
16 - 22 (4)
21 - 22 (3)
31 - 22 (2)
31 - 56 (1)

1)

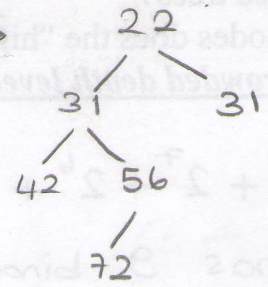



(leftist)

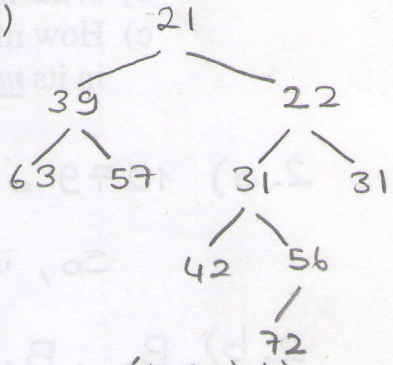
2)




(not leftist)

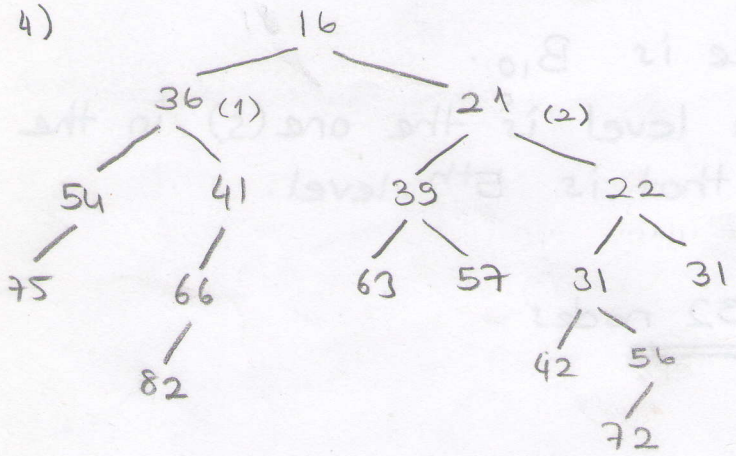



3)

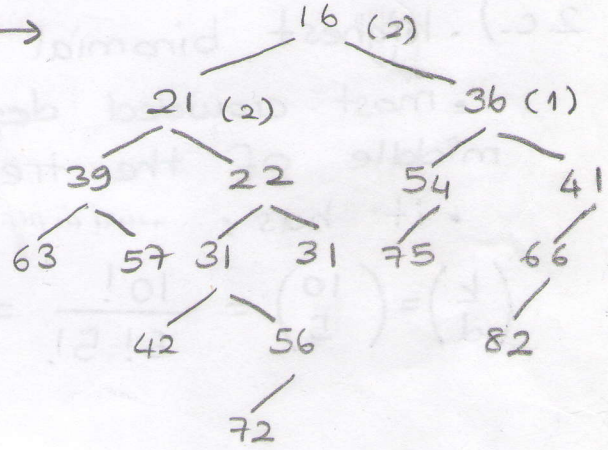



(leftist)

4)

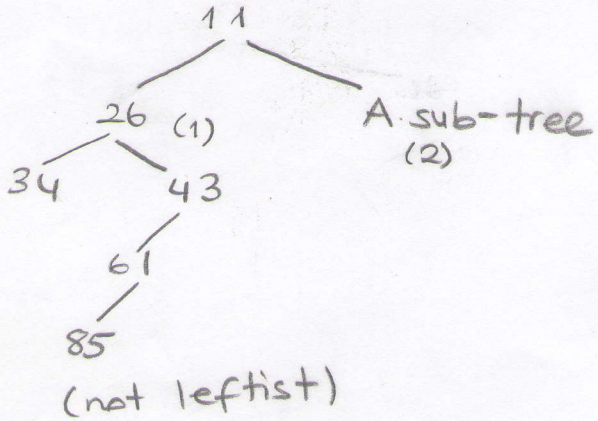



(not leftist)

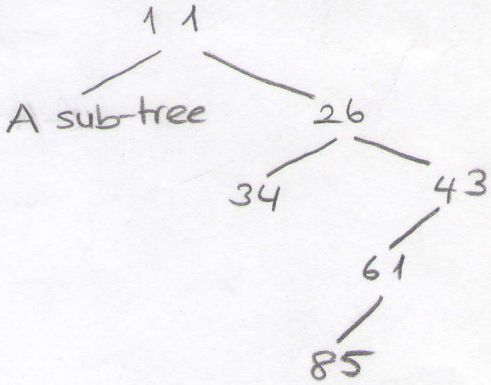



(A sub-tree)

5)




(not leftist)




(result)